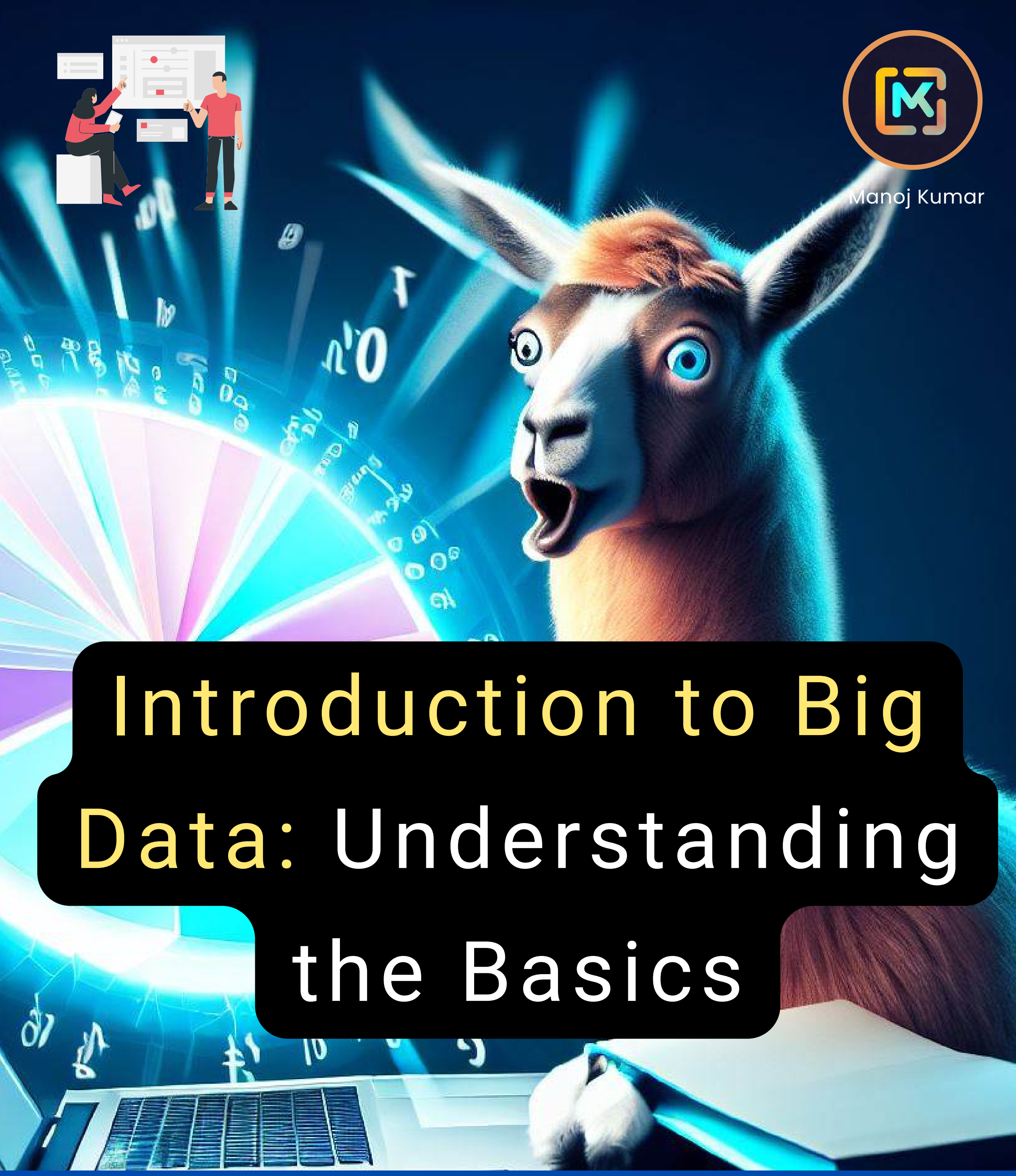

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Introduction to Big Data: Understanding the Basics

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Big Data

Big data refers to large and complex datasets that are too big for traditional databases to handle.

These datasets come from various sources and formats, such as audio, video, photos, websites, social media, and more. Big data can help you understand your products and services better, generate valuable insights, and make smarter decisions.

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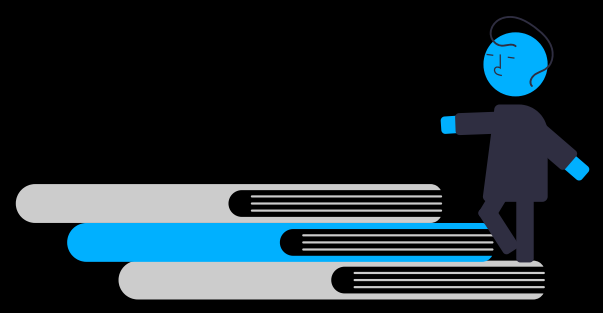

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Types of Big Data

There are different types of big data that require different tools and methods. Here are some examples:

- **Structured data**: This is highly organized data that follows a specific format. It is easy to retrieve and includes information like mobile numbers, employee details, and salaries. Data stored in relational databases is an example of structured data.
- Unstructured data: This is highly unorganized data that does not follow a specific format. It cannot be stored in relational databases and needs to be transformed into a structured format before analysis. 80% of data in an organization is unstructured, and it comes in multiple formats such as images, audio, video, social media posts, and more.
- Semi-structured data: This is a combination of structured and unstructured data that has some identifying characteristics but no fixed structure. For example, images may have metadata or tags, but the information within has no structure. XML or JSON files are examples of semi-structured data.

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5 V's in Big Data

To handle big data effectively, you need to consider the 5 V's:

- Volume: The massive amount of data that is growing exponentially. The data can range from terabytes to petabytes and beyond. You need distributed systems like Hadoop and MongoDB to store such large amounts of data.
- Velocity: The speed at which data is generated, stored, and analyzed. The velocity of data generation is especially high in real-time applications such as social media, which produces audio, videos, posts, etc. at a rapid pace.
- Variety: The different types of data that are used daily. You need technologies that can handle different data formats (text, audio, videos, etc.) and integrate them for analysis.
- Veracity: The quality or trustworthiness of the available data. You need to ensure the accuracy and reliability of the data you analyze, especially in real-time scenarios where the data can be subject to errors, typos, and other inaccuracies.
- Value: The need to convert raw data into something valuable and useful. You need to extract meaningful information that results in a positive return on investment.

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Use Case : Netflix

Netflix analyzes our data about what we're watching or searching, extracting the data points from that, like what titles customers watch, what genre they like, how often playback stopped, ratings are given, etc., and feeding that to its recommendations system. This will make decisions smooth and firm in terms of knowing the customers' needs rather than assuming them (what most companies do). The major data structures used in this process include Hadoop, Hive, Pig, and other traditional business intelligence

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Top Tools in Big Data

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Big data is not just a buzzword. It is a revolution that can transform your business in 2023 and beyond.

- Are you ready to embrace big data?
- What are some of the challenges you face with big data?
- How do you use big data in your daily work?

Let me know in the comments below.

And if you found this post helpful, please like, share, and follow me for more content on big data and other topics.

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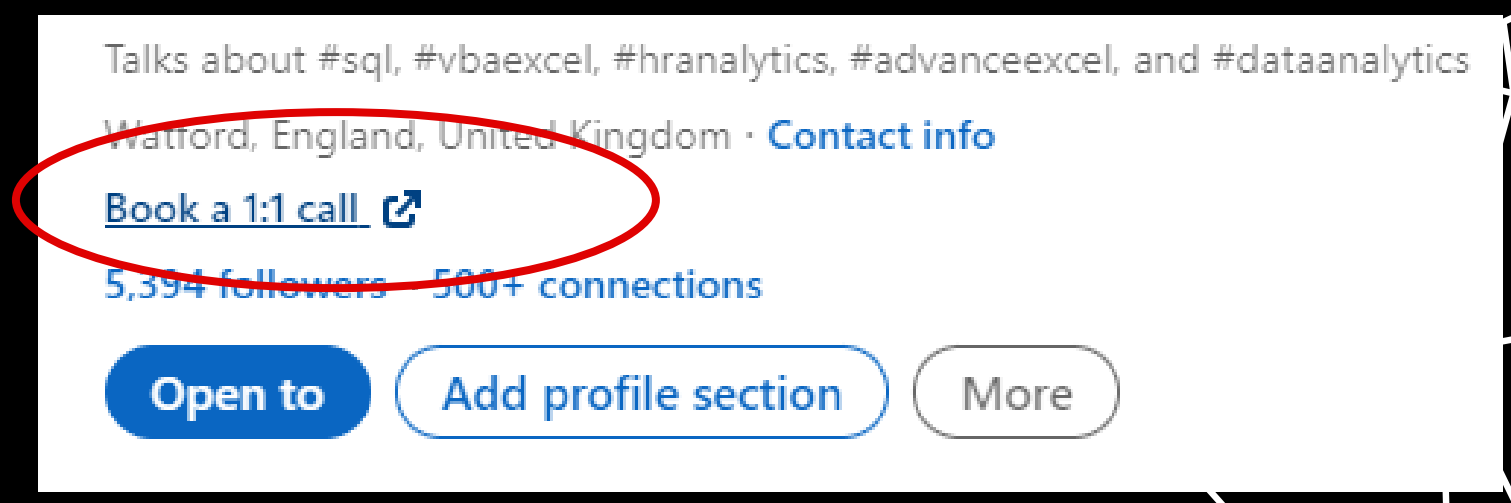

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